

ModulUS: A Sandbox for High-Resolution Wearable Ultrasound Development

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Abstract—We introduce *ModulUS*, a modular sandbox platform for wearable ultrasound (US) development. The four-board system integrates a 32-channel pulser, an analog frontend (AFE) with bandwidth reduction, and an ARM Cortex-M4 microcontroller (MCU) with a dual 5 Msps, 12 bit analog-digital converter (ADC), each implemented as a swappable module to support rapid prototyping and evaluation of alternative circuit architectures. To demonstrate its functionality, we validated the AFE board, which reduces the effective signal bandwidth by a factor of 4 while preserving structural features, with envelope fidelity confirmed against digital Hilbert transforms ($R^2 = 0.88$). Compared to State-of-the-Art (SoA) platforms, *ModulUS*-derived designs set a new benchmark in power efficiency, achieving the lowest reported average power normalized by excitation frequency (0.6 mW/MHz), down to two orders of magnitude below existing MCU- and Field Programmable Gate Arrays (FPGA)-based systems. By shifting bandwidth reduction (BWR) into the analog domain, *ModulUS* enables high-frequency, low-power US acquisition and establishes a flexible, reconfigurable testbed for next-generation wearable US technologies.

Index Terms—Wearable Ultrasound, Modular Design, Bandwidth Reduction, Analog Front-End, Low-Power Ultrasound, Microcontroller-Based, High-Frequency Ultrasound

I. INTRODUCTION

A growing trend in wearable US system design is the shift from conventional medical imaging towards the acquisition, processing, and machine interpretation of raw US data directly on digital probes [1]. Emerging non-imaging US wearables are typically optimized for recording A-mode or M-mode echoes, often via distributed transducer networks [2], and leverage machine learning (ML) for data interpretation, thereby reducing the traditionally high operator dependence of US.

From a system architecture perspective, two main templates (see Tab. I) dominate the field: (a) medium-channel-count wearable platforms with up to 32 channels implemented on FPGA-based architectures, and (b) low-channel-count platforms (1–8 channels, often partially time-multiplexed) employing MCU-based topologies. The primary differences between these architectures lie in their achievable axial spatial resolutions (i.e., the maximum transducer excitation frequencies), temporal resolutions (i.e., the maximum pulse-repetition frequencies), number of parallel receive (Rx) channels, and trade-offs in power consumption and form factor (i.e., volume).

FPGA-based wearables [5] are, in principle, fully capable US systems that can support standard medical modalities such as A- and M-mode scans, B-mode imaging, and Doppler,

Wearable US Systems	Topology	n_{Rx}	Volume [cm ³]	Power/ n_{Rx} [mW]	Power/ $f_{Tx}n_{Rx}$ [mW/MHz]
WULPUS [3]	MCU	1	*14.95	*22	5.5
USoP [4]	MCU	1	*16.2	614	102.3
TinyProbe [5]	FPGA	32	39.9	30.3	2.0
[†] PuLsE [6]	MCU	1	‡12.6	5.8	↓ 0.6

n_{Rx} denotes the number of Rx channels, and f_{Tx} denotes the excitation frequency of the transducer. Reported power metrics were extracted from the cited literature and, where specified, correspond to pulse repetition frequencies of at least 25 Hz.

* System size and power values were reported in [5].

[†] This system was developed on *ModulUS*.

‡ The referenced paper only stated that the system can be integrated into a smartwatch; for completeness we assumed a diameter of 40 mm and a thickness of 10 mm, corresponding to commercial smartwatch form factors [7].

↓ Indicates the best (lowest) value among the compared systems.

TABLE I

COMPARISON OF WEARABLE US SYSTEMS, INCLUDING PROCESSING TOPOLOGY, NUMBER OF RX CHANNELS, DEVICE VOLUME, AND POWER CONSUMPTION. REPORTED POWER IS ADDITIONALLY NORMALIZED PER RX CHANNEL AND TO THE EXCITATION FREQUENCY (f_{Tx}) OF THE TRANSDUCER.

but at the cost of higher power consumption and larger form factor compared with MCU-based wearables [6] (see Tab. I). Conversely, MCU-based platforms are substantially smaller and more power-efficient, but significantly constrained in spatial and temporal resolution [3].

For many MCU-based US wearables without imaging capabilities, hardware constraints make it impractical or even impossible to capture modalities in which phase information is critical (e.g., beamforming, fast displacement tracking, or Doppler) [8]. Even when radio frequency (RF) data are recorded and digitized, phase information is often discarded during subsequent digital signal processing [9], [10]. Supporting phase information typically requires full-bandwidth ADC, which imposes substantial computational overhead, often exceeding $5\times$ that of bandwidth reduced methods [11]. To address this limitation, analog signal-conditioning approaches for wearable US have been proposed [12], which reduce the effective bandwidth of the received signals before digitization while preserving echo resolution that is set by the transmit signal bandwidth. By easing the digitization burden, these methods can expand the capabilities of MCU-based wearable US systems, relaxing existing hardware requirements and enabling higher-frequency, higher-resolution applications.

Despite these recent advances, developing a wearable US system remains challenging, requiring deep mixed-signal expertise and cross-disciplinary domain knowledge. Most open

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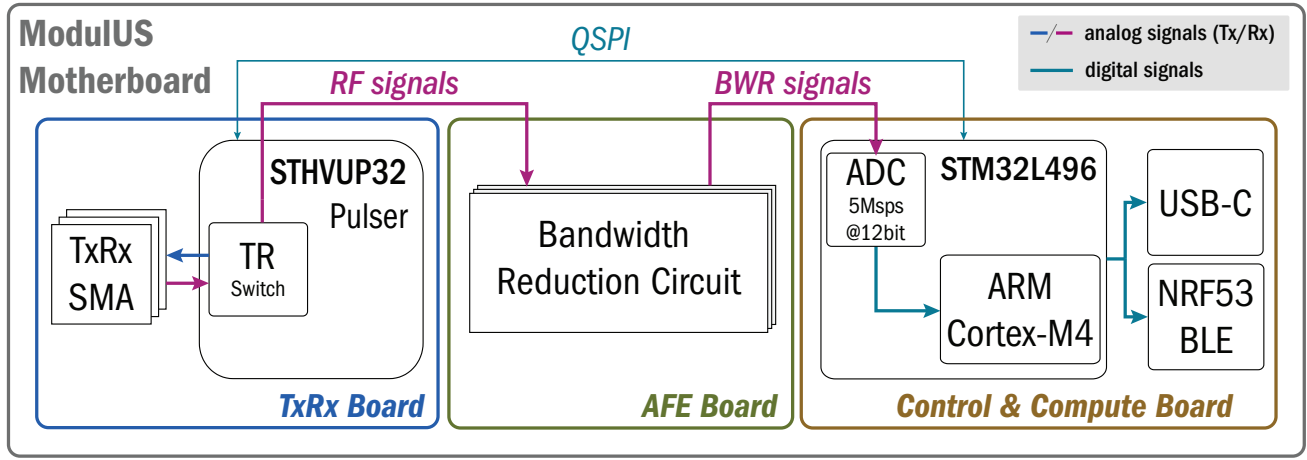


Fig. 1. High-level circuit block diagram of the *ModulUS* platform. The system integrates a control and compute board (MCU, ADC, memory, and communication interfaces), an analog front-end board for signal conditioning and bandwidth reduction, and a Tx/Rx board with pulser, switching, and transducer interfacing.

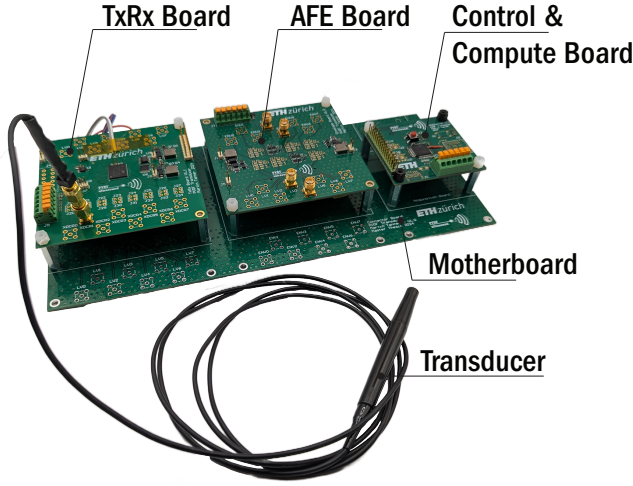


Fig. 2. Photograph of the assembled *ModulUS* platform showing the motherboard with its functional boards.

research systems use FPGA-based designs [13], which are complex to develop and adapt, whereas open-source MCU-based architectures are well documented and straightforward to use [14] but lack the flexibility to modify modules such as the AFE, for evaluating new BWR techniques due to their high level of integration.

In this context, we present *ModulUS*, a modular and flexible hardware sandbox for wearable US development. The primary contribution of this work is a detailed description of the platform's design concept and a demonstration of its functionality, showing how it employs an AFE processing board to preserve high axial resolution with a 10 MHz transducer while reducing conversion, sequencing, and digital computing requirements. This design enables data acquisition without complex high-speed interconnects to ADCs (e.g., LVDS or JESD204), instead leveraging an ARM Cortex-M4 MCU with advanced analog features and an integrated 5 Mps ADC.

II. SYSTEM DESIGN

A. Architecture

The *ModulUS* system architecture (Fig. 1) is designed for modularity, allowing flexible modification of the core building blocks of a wearable US platform. These include the Tx and Rx stages, the AFE, and the control and compute unit. The complete system (Fig. 2) consists of four functional boards and measures 260 x 115 x 40 mm:

- 1) The TxRx board integrates a 32-channel US pulser (STHVUP32, STMicroelectronics, Switzerland), with 8 exposed channels in Tx and Rx,
- 2) The AFE board integrates the BWR circuits for all channels, as described in [11], enabling envelope detection,
- 3) The control and compute board is based on an STM32L496 MCU (STMicroelectronics, Switzerland), featuring a dual 5 Mps, 12 bit on-chip ADC. It provides a USB-C interface for data transfer either via Universal Asynchronous Receiver/Transmitter (UART) or through an NRF53 Bluetooth low energy (BLE)-capable MCU to a connected subsystem,
- 4) The *ModulUS* motherboard interconnects all daughter modules using high-density board-to-board connectors.

B. Functionality

The pulser is configured by the MCU via a Quad Serial Peripheral Interface (QSPI) interface, which programs its sequencing logic. After boot, the MCU initializes the pulser and starts pulse generation, transmitting signals to the transducer. The returning echoes pass through the transmit-receive switch (TR) to the AFE board, where envelope detection is applied. For each received US pulse, an interrupt triggers an ADC capture on the MCU. The acquired data is then processed, stored locally, or transmitted to the host system. Power is supplied to each board via dedicated edge connectors, ensuring electrical isolation between subsystems and enabling flexible reconfiguration for experimental setups.

C. Figure of Merit Comparison

To compare wearable US systems with different transducer center frequencies (f_{TX}), we define the average power per Rx channel normalized by f_{TX} (mW/MHz) as a figure of merit. Since higher frequencies improve axial resolution but proportionally increase the energy required for signal conversion, storage, computation, and transfer [15], normalization provides a resolution-aware basis for comparing MCU- and FPGA-based architectures with the results presented in Table I and discussed in Section IV.

III. EXPERIMENTAL TESTING

A. Setup

The system was validated using a single-channel 10 MHz transducer (Piezohannas, China) immersed in water. The transducer emitted five-cycle pulses ($n = 5$) at 15 V_{pp} toward a 0.7 mm polyethylene terephthalate (PET) disc, which served as a reflecting interface. Both raw Rx echoes and processed analog envelopes, generated by the AFE module, were acquired simultaneously on an acquisition card (ADQ412, Tele-dyn Technologies, USA).

Axial resolution was approximated as half the spatial pulse length (SPL) [16], with $SPL = n\lambda$. Using a sound speed of $c = 1540 \text{ m s}^{-1}$ and a center frequency of $f = 10 \text{ MHz}$, the wavelength is $\lambda = c/f = 0.154 \text{ mm}$. This yields $SPL = 0.77 \text{ mm}$ for $n = 5$ cycles and an axial resolution of about 0.38 mm, sufficient to resolve the two surfaces (upper and lower boundaries) of the 0.7 mm disc.

The minimum frequency required to resolve a target of thickness d follows from the resolution criterion $R_{ax} \leq d$, which gives

$$f_{\min} = \frac{nc}{2d}. \quad (1)$$

For $n = 5$, $c = 1540 \text{ m s}^{-1}$, and $d = 0.7 \text{ mm}$, this evaluates to $f_{\min} \approx 5.5 \text{ MHz}$, indicating that probes below this frequency cannot resolve the disc under comparable pulse conditions.

B. Evaluation

Two metrics were used to assess the performance of the *ModulUS* AFE. First, Welch power spectral densities were computed for the raw Rx signals and the corresponding BWR envelopes. The mean spectral frequency was calculated as

$$f_{\text{mean}} = \frac{\sum_i f_i P(f_i)}{\sum_i P(f_i)}, \quad (2)$$

where f_i denotes the frequency bins and $P(f_i)$ the spectral power at each frequency. The ratio of mean frequencies quantified the effective bandwidth reduction. Second, the fidelity of the analog envelopes was compared to digitally derived envelopes obtained via the Hilbert transform. Prior to comparison, signals were corrected for DC bias and filter delay. The coefficient of determination was computed as

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad (3)$$

where y_i are the analog envelope samples, $\hat{y}_i$ the corresponding Hilbert-derived samples, and $\bar{y}$ their mean.

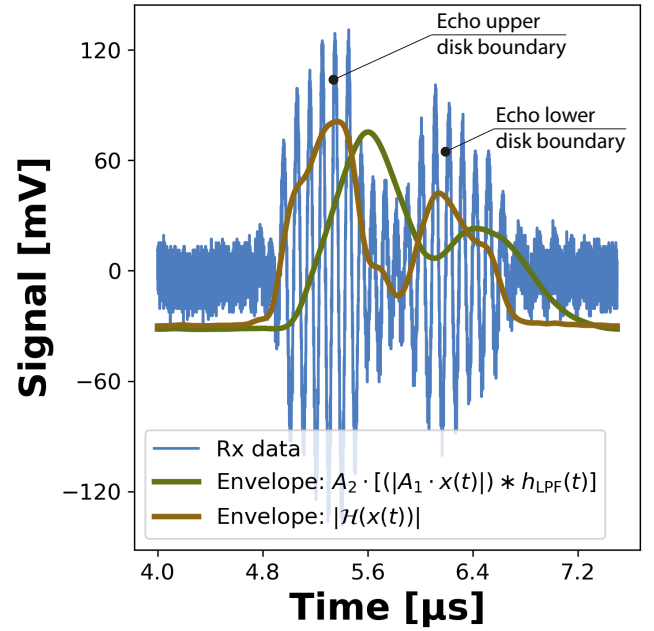


Fig. 3. Validation results using a 10 MHz transducer and a 0.7 mm PET disc. The raw receive (Rx) signal (blue), the analog envelope obtained from the analog front-end ($A_2 [|A_1 x(t)| * h_{LPF}(t)]$) (green), and the digitally derived Hilbert envelope ($|H(x(t))|$) (brown) are shown. Here, A_1 denotes the prerectification gain applied to the RF signal, $h_{LPF}(t)$ is the impulse response of the low-pass filter used to extract the envelope (which introduces a group delay relative to the Hilbert method), and A_2 represents post-filter gain. The analog and digital envelopes exhibit close correspondence ($R^2 = 0.88$), and the PET disc generates two resolvable echo peaks despite an approximately fourfold bandwidth reduction.

C. Results

The mean frequency of the raw Rx signals was approximately 2.9 MHz, while the BWR envelopes exhibited a mean frequency of 0.7 MHz, corresponding to a reduction by a factor of about four. The PET disc produced two resolvable echo peaks in the BWR envelopes, demonstrating that essential structural information was preserved despite the reduced bandwidth. Moreover, analog and digitally derived envelopes showed close correspondence with $R^2 = 0.88$, with differences limited to the expected phase removal. The approximately fourfold bandwidth reduction enabled direct digitization using the on-chip ADC of the MCU, eliminating the need for high-speed FPGA-based acquisition. Fig. 3 illustrates representative results, showing the raw Rx signal, the analog envelope obtained from the AFE, and the digitally derived envelope for direct comparison.

IV. DISCUSSION

The results confirm that *ModulUS* achieves sub-millimeter axial resolution while operating entirely on a low-power MCU. This capability is enabled by the analog BWR frontend, which compresses the effective signal spectrum by a factor of four without erasing structural information. When positioned against the SoA (Tab. I), platforms based on the *ModulUS* concept (e.g., *PuLsE* [6]) show a distinct trade-off profile.

FPGA-based systems such as *TinyProbe* [5] offer superior flexibility, with up to 32 channels and native support for

imaging and advanced US modalities (e.g., Doppler). However, supporting multiple channels with high spatio-temporal resolutions comes at the cost of increased power consumption (970 mW) and a substantially larger volume (39.9 cm³) compared to MCU-based single channel systems. In contrast, MCU-based solutions such as *WULPUS* [3], *USoP* [4], and *PuLsE* [6] achieve much lower power and volume but remain fundamentally limited to single-channel operation.

To enable a fair comparison across architectures operating at different transducer frequencies, we introduced the normalized average power per Rx channel (mW/MHz) as a figure of merit (see Section II-C). Using this metric, *ModulUS*-derived systems such as *PuLsE* [6] set a new benchmark with 0.6 mW/MHz, outperforming all previously reported wearable designs (Tab. I). For reference, *USoP* consumes 102.3 mW/MHz, *TinyProbe* 2.0 mW/MHz, and *WULPUS* 5.5 mW/MHz. This efficiency gain, exceeding two orders of magnitude compared to *USoP*, and remaining 3–10x lower than *TinyProbe* and *WULPUS*, demonstrates that envelope-domain acquisition with analog BWR can decouple high-resolution imaging from excessive energy budgets.

These results have important implications. First, *ModulUS* demonstrates that MCU-based platforms are not strictly limited to low-frequency or low-resolution applications. By pushing BWR into the analog domain, *ModulUS* enables operation of 10 MHz transducers with power figures that surpass other ultra-low-power MCU solutions working at much lower frequencies. Second, the modular design of *ModulUS* opens up a level of reconfigurability for the development of individual wearable system designs. Unlike highly integrated devices, *ModulUS* exposes the AFE, pulser, and control logic as interchangeable components, lowering the barrier for future research in alternative BWR schemes or wearable transducer development.

Naturally, these gains come with limitations. *ModulUS* processes only one channel at a time; this rules out simultaneous multi-channel imaging. Furthermore, the envelope-domain path discards phase information, meaning it cannot directly support applications such as Doppler or displacement tracking. Finally, *ModulUS* is not a wearable US system per se but a catalyst. Its modularity and flexibility come at the cost of size and weight, positioning it as a sandbox for wearable US development rather than a deployable device.

V. CONCLUSION

We introduced *ModulUS*, a four-board sandbox for developing high-resolution wearable US systems. By shifting BWR into the analog domain, *ModulUS* moves the performance bottleneck away from digital acquisition, enabling sub-millimeter resolution on low-power MCU platforms. The system preserves structural information at 10 MHz while reducing bandwidth more than fourfold. Its modular design, exposing pulser, AFE, as well as control and compute units as interchangeable modules, provides a versatile platform for wearable US research. These gains entail trade-offs. Compared to higher-power FPGA-based systems, *ModulUS* is

limited to single-channel operation, restricting simultaneous 2-D imaging. Furthermore, its envelope-based BWR omits phase information, required by many modalities such as delay-and-sum beamforming, Doppler, or displacement tracking. Addressing this would involve phase-preserving strategies (e.g., I/Q sampling or alternative BWR methods), at the cost of higher power and system complexity. Future work will focus on refining *ModulUS* through standardized module form factors, comprehensive in vivo validation, and open-sourcing the hardware design.

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